

Machine Learning

2025 Fall

Assignment 3

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1. Reading and Explaining Lemmas

Your task is to read the following paper :

Ryck et al., On the approximation of functions by tanh neural networks.

Focus on Lemma 3.1 and Lemma 3.2.



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Neural Networks

journal homepage: www.elsevier.com/locate/neunet



On the approximation of functions by tanh neural networks

Tim De Ryck*, Samuel Lanthaler, Siddhartha Mishra

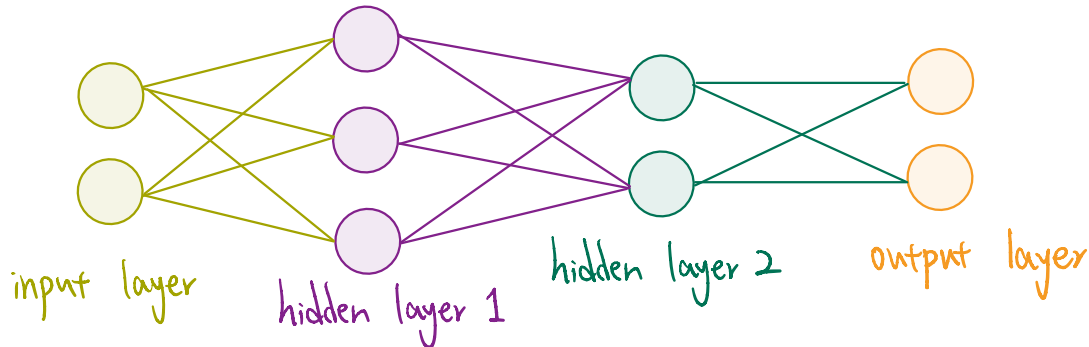
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Background knowledge

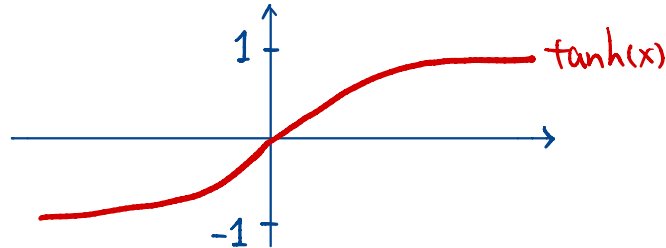
A **neural network** is composed of an input layer, one or more hidden layers, and an output layer.

Each "neuron" applies weights to inputs, sums them, and passes the result through an **activation function**, allowing the network to learn non-linear relationship.



Background knowledge

The **tanh function**, $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, outputs values between -1 and 1 and is symmetric about zero.



A **shallow tanh neural network** has just one hidden layer between the input and output layer and uses the tanh activation function.

Background knowledge

The **width** of a neural network is defined as the maximal number of neurons in any single layer.

For example, Fig 1 shows a neural network of width 3.

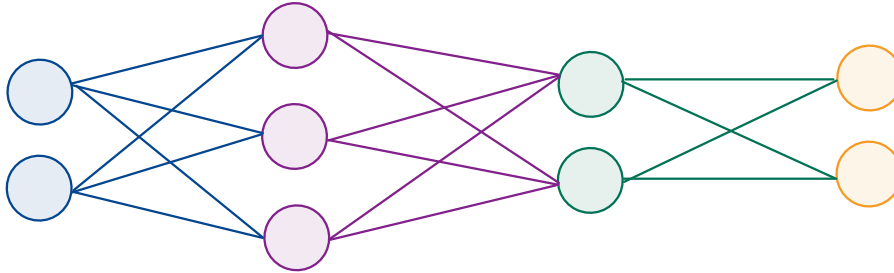


Fig 1

Background knowledge

We define the p th order central finite difference operator S_h^p for the tanh activation function σ for some $p \in \mathbb{N}$ by

$$S_h^p[\sigma](x) = \sum_{i=0}^p (-1)^i \binom{p}{i} \sigma\left(x + \left(\frac{p}{2} - i\right)h\right).$$

Next we define for any $p \in \mathbb{N}$, $q \in 2\mathbb{N}-1$ and $M > 0$ the monomial $f_p: [-M, M] \rightarrow \mathbb{R}$ and the tanh neural networks $\hat{f}_{q,h}: [-M, M] \rightarrow \mathbb{R}$ as

$$f_p(y) := y^p \quad \text{and} \quad \hat{f}_{q,h}(y) := \frac{S_{h,y}^q[\sigma](0)}{\sigma^{(q)}(0) h^q}$$

Background knowledge

For example, $f_5(2) = 2^5 = 32$.

$$\begin{aligned}\hat{f}_{5,0.01}(2) &= \frac{\mathcal{D}_{0.02}^5[\sigma](0)}{\sigma^{(5)}(0)(0.01)^5} \\ &= \frac{\sum_{\lambda=0}^5 (-1)^\lambda \binom{5}{\lambda} \sigma\left(\left(\frac{5}{2} - \lambda\right) \times 0.02\right)}{16 \times (0.01)^5} \\ &\approx 31.95\end{aligned}$$

Note that $\hat{f}_{5,0.01}(2) \approx f_5(2)$.

Lemma 3.1.

Let $k \in \mathbb{N}_0$ and $s \in 2\mathbb{N} - 1$. Then it holds that for all $\varepsilon > 0$ there exists a shallow tanh neural network $\Psi_{s,\varepsilon} : [-M, M] \rightarrow \mathbb{R}^{\frac{s+1}{2}}$ of width $\frac{s+1}{2}$ such that

$$\max_{p \leq s, p \text{ odd}} \|f_p - (\Psi_{s,\varepsilon})_{\frac{p+1}{2}}\|_{W^{k,\infty}} \leq \varepsilon.$$

Moreover, the weights of $\Psi_{s,\varepsilon}$ scale as

$$O(\varepsilon^{-\frac{s}{2}} (2(s+2)\sqrt{2M})^{s(s+3)}) \text{ for small } \varepsilon \text{ and large } s.$$

Explanation

Lemma 3.1 tells us that for any monomial of odd degree S (like $x, x^3, x^5, \dots$), there exists a shallow tanh neural network of width $\frac{S+1}{2}$ that can approximate that monomial as closely as we want — not just in terms of function values, but also considering a measure of smoothness called the Sobolev norm (which takes derivatives into account).

Idea

As we see in the previous example that $\hat{f}_{5,0.01}(2) \approx f_5(2)$, we may guess $\hat{f}_{p,h} \approx f_p$.

Actually, by Taylor's theorem and some computation, we can prove for arbitrary $k \in \mathbb{N}$:

$$\|f_p - \hat{f}_{p,h}\|_{W^{k,\infty}} \leq ((2(p+2)pM)^{p+3} + (2pk)^{k+1})h^2 =: \varepsilon.$$

Recall that $\hat{f}_{p,h}(y) := \frac{\delta_{h,y}^p[\sigma](0)}{\sigma^{(p)}(0)h^p}$, where $\sigma(x) = \tanh(x)$

is our activation function, thus we get the shallow tanh neural network naturally.

Lemma 3.2.

Let $k \in \mathbb{N}_0$, $s \in 2\mathbb{N} - 1$ and $M > 0$. For every $\varepsilon > 0$, there exists a shallow tanh neural network

$\psi_{s,\varepsilon} : [-M, M] \rightarrow \mathbb{R}^s$ of width $\frac{3(s+1)}{2}$ such that

$$\max_{p \leq s} \|f_p - (\psi_{s,\varepsilon})_p\|_{W^{k,\infty}} \leq \varepsilon.$$

Furthermore, the weights scale as $O(\varepsilon^{-\frac{s}{2}} (\sqrt{M}(s+2))^{3s(s+3)/2})$ for small ε and large s .

Explanation

Lemma 3.2 extends the previous result to monomials of even degree S (like $x^2, x^4, x^6, \dots$). It guarantees that there exists a shallow tanh neural network of width $\frac{3(S+1)}{2}$ that can approximate the monomial as closely as we want — not just in terms of function values, but also considering a measure of smoothness called the Sobolev norm (which takes derivatives into account).

Idea

Since for any even number n , we have

$$\frac{(x+1)^{n+1} - (x-1)^{n+1}}{2} = \binom{n+1}{1} x^n + \binom{n+1}{3} x^{n-2} + \dots + \binom{n+1}{n+1} x^0,$$

that is, we can construct the even power monomial x^n from the odd power monomials $x, x^3, \dots, x^{n-1}, x^{n+1}$.

Thus, by Lemma 3.1 and induction, we can get the shallow tanh neural network naturally.

Reference

De-Ryck T, Lanthaler S, Mishra S. On the approximation of functions by tanh neural networks. Neural Netw. 2021; 143: 732-50.